Do NHL Players Outperform in Contract Years? An Empirical Analysis

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Abstract

This paper investigates whether NHL players exhibit statistically significant improvements in performance during contract years, utilizing a panel dataset spanning 2008 to 2024. We test the contract year hypothesis using fixed-effects regressions and various performance metrics, including goals, assists, and advanced analytics. We find that through our empirical model, that both goalies and skaters do not experience significant dips or increases in their performance. This contradicts answers found in other literature.

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1 Introduction

Since the early 2000s, all four major sports leagues the NHL, NBA, NFL, and MLB have put some kind of salary control in place. The NHL and NFL use a hard cap, the NBA uses a soft cap, and the MLB has a luxury tax system. In our project, we look at the NHL and try to answer a simple question: do players perform better in a contract year? That means the last season of their current deal before they become free agents.

The thinking here is pretty basic. If there's money on the line, players might be more motivated to boost their stats and play at a higher level. But once they sign a new contract, especially if it's long and guaranteed, that extra motivation might disappear. This idea fits well with economic incentive theory. People respond to rewards, and athletes are no different.

We can also look at it from a team perspective. NHL teams have a limited amount of money they're allowed to spend because of the hard salary cap. At the same time, they want as many high-performing players as possible. So when a player does really well in a contract year while still being paid under their old deal, it forces teams to use more of their limited cap space than expected. That creates a tough balance between wanting good players and staying under the cap.

We focus on the NHL in part because it has received less attention in the contract year literature compared to leagues like the NBA, NFL, or MLB. While prior studies often examine how contract incentives affect scoring, playing time, or salary outcomes in those leagues, the NHL remains relatively understudied, particularly in terms of distinguishing between skaters and goalies, as well as unrestricted versus restricted free agents. This gap presents an opportunity to explore how contract structures and performance incentives operate in a different competitive and bargaining environment.

We have also seen some pretty questionable contracts handed out in recent years. Big name players like Elias Pettersson, Jeff Skinner, and Elias Lindholm have all signed massive deals and

then underperformed not long after. While this can be frustrating for fans, it also has a growing economic impact. There's more money in professional sports now than ever before. We've seen a huge influx of dollars into the NHL through betting markets, skyrocketing TV deals, and rising ticket prices. All of this adds more financial value to the on ice product, and when a team gives out a bloated contract that doesn't live up to expectations, it can hurt that value. Overpaying players isn't just a roster problem it's an economic one too.

We're using NHL performance data from the NHL API and MoneyPuck, along with contract info from Spotrac, to build a panel dataset that tracks players over multiple seasons. This should give us a good way to test if players play better when they're trying to earn their next contract. To see why the NHL makes sense for this kind of question, it helps to look at how contracts work under the salary cap system.

Section II gives background on the NHL and reviews past research on contract year performance. Section III introduces our empirical model and explains why we chose it. Section IV covers our dataset, including where we got the data and how we cleaned it. Finally, Section V presents our findings.

Through our research, we find that the media and analysts often overreact to a few standout cases and overlook the broader data. While certain players may have had strong contract years, our results show that, on average, players do not outperform their career norms in the final year of a contract. Using an empirical model, we tested a binary variable indicating whether a player was in a contract year and found no statistically significant difference in performance. This suggests that the contract year effect, while popular in narratives, does not hold up across the league.

2 Background

2.1 General Information

The National Hockey League (NHL) is the premier professional hockey league in the world and represents the highest level of competition among North America's five major sports leagues. It

consists of 32 teams, 25 based in the United States and 7 in Canada. Each team competes in an 82-game regular season from October to mid-April, after which the top-performing teams advance to the playoffs. The postseason culminates in the Stanley Cup Final, a best-of-seven series between the Eastern and Western Conference champions, with the winner crowned as league champion.

Player performance in the NHL is typically evaluated based on how much an individual contributes to team success. For skaters (all non-goalies), one of the most commonly used metrics is points, calculated as the sum of goals and assists. This remains a core measure regardless of position, whether winger, center, or defenseman. However, the use of advanced statistics has expanded in recent years. Metrics such as expected goals (xG), shot quality, and other context-sensitive measures aim to capture a player's on-ice contributions more accurately than traditional counting stats. These metrics often play a role in contract negotiations. Despite the growing influence of analytics, conventional measures and subjective evaluations, often referred to as the "eye test", continue to influence public perception and management decisions. If a player appears effective on the ice and is generating points, they are likely to attract positive attention.

Career trajectories in the NHL vary widely. Some players begin their careers at 18 or 19 years old and continue playing into their late 30s. Others have shorter, less stable careers due to factors like injury, inconsistent performance, or limited opportunity. The structure and timing of contracts are largely governed by the league's Collective Bargaining Agreement (CBA), which dictates conditions such as the length of entry-level deals, eligibility for free agency, and the process of salary arbitration. These institutional rules shape how players are evaluated and compensated throughout their careers.

A particularly relevant concept within this system is the "contract year," defined as the final season of a player's existing deal. It is widely believed, by fans, analysts, and sometimes even coaches, that players increase their effort and production in these seasons to maximize their value heading into negotiations. The rationale is intuitive: if future earnings depend directly on current performance, players have a financial incentive to elevate their play. This phenomenon is known as the "contract year effect," and this paper investigates whether such an effect is observable in the

NHL.

2.2 The NHL Salary Cap and Contract Structure

Since the 2005–06 season, the NHL has operated under a hard salary cap system. This imposes an upper limit on team spending for player salaries, which adjusts annually based on league revenues. The cap creates significant strategic constraints: teams must balance spending across the roster, making efficient use of cap space critical. As a result, the value and timing of player contracts have become increasingly important in team-building decisions. Unlike some other leagues where a portion of salary may be tied to performance incentives, NHL contracts are fully guaranteed in nearly all cases. This means that players receive their full salary regardless of their on-ice performance. While performance bonuses do exist, they are typically reserved for players on entry-level contracts or veterans on short-term deals and tend to be relatively minor in scope. For the majority of NHL players, compensation is predetermined and not directly influenced by statistical output.

Contract lengths vary, ranging from one- or two-year bridge deals to long-term extensions that span up to eight years, the maximum allowed under the current CBA. Teams may prefer shorter contracts for flexibility or longer ones for cost certainty. Importantly, the absence of large-scale performance-based incentives means that players do not automatically earn more money by exceeding expectations in a given season.

One mechanism that links performance and pay is salary arbitration. Restricted Free Agents (RFAs) who meet certain criteria can request arbitration, where a neutral third party determines a fair salary based on recent performance, career progression, and comparable players. This system incentivizes strong performance in the lead-up to arbitration-eligible seasons, as statistical output plays a direct role in salary determination. Therefore, the existence of arbitration enhances the potential relevance of contract years, particularly for players who have not yet reached unrestricted free agency.

Overall, the interplay between the hard salary cap, the prevalence of guaranteed salaries, and

the structured arbitration process creates a context in which both teams and players face strategic decisions regarding contract timing and performance. For players, especially those approaching the end of their current deals, there may be greater motivation to enhance performance in order to secure future earnings. This makes the NHL a compelling setting to test the contract year effect through an economic lens.

2.3 Literature Review

A foundational concept in labor economics is that workers respond to incentives. Just as a student is more likely to study harder for a math test that counts for 40

2.3.1 Contract Year Effects

A central question in this literature is whether players increase effort and performance during the final year of their contracts, and conversely, whether they reduce effort after securing long-term deals. Frick (2011), studying German soccer, finds that players perform better and more consistently in contract years, as reflected in subjective evaluations from Kicker magazine. The narrowing of performance variance suggests not only harder work but also more focused play during periods of financial uncertainty.

Stiroh (2006) presents similar findings in the NBA, showing that team success improves when more players are in contract years, but declines when rosters are dominated by recently re-signed players. These patterns support the presence of moral hazard, a key concept in economics that describes how agents may reduce effort when they are insulated from consequences. In this case, guaranteed long-term contracts may reduce the incentive to maintain peak performance once financial security is attained.

Maxcy et al. (2002) examine this dynamic in MLB and directly test the shirking hypothesis, the idea that players exert less effort after securing a contract. Their regression models show increased performance leading up to contract negotiations but find little evidence of effort decline afterward. This suggests that external factors, such as monitoring, performance bonuses, and public account-

ability, may help deter shirking in some settings.

2.3.2 Post-Contract Performance Decline

Concerns about post-contract effort reductions are closely tied to the principal-agent problem: teams (principals) rely on players (agents) to exert effort, but cannot perfectly monitor behavior. Guaranteed multi-year deals can create a misalignment of incentives, where players may no longer be motivated to maintain high performance once their future salary is secure. Stiroh (2006) documents this effect in the NBA, where performance gains during contract years are often followed by statistically significant declines post-signing. These changes have implications not just for individuals, but also for team-level outcomes, including win rates.

Despite these concerns, teams continue to offer long-term contracts. Stiroh explains this by pointing to relationship-specific investments, injury protection, and bargaining frictions, all of which make short-term contracts impractical. Consequently, moral hazard becomes a systemic risk shaped by the institutional structure of sports labor markets. In contrast, Maxcy et al. (2002) find that in MLB, performance bonuses and public scrutiny may offset these risks, reducing the occurrence of post-contract shirking. These conflicting findings highlight important differences across sports in how incentives are structured and monitored.

2.3.3 Team Dynamics and Wage Equity

Another relevant topic in the sports labor literature concerns how wage structures within teams affect morale and performance. DeBrock et al. (2004) analyze this issue in MLB, finding that larger salary disparities among teammates, especially those not aligned with actual performance, correlate with worse team outcomes. This supports the argument that unexplained pay gaps can undermine cohesion, reduce morale, and hurt team productivity. Their study identifies two key mechanisms. First, teams that pay above market expectations often benefit from higher individual output. Second, flatter wage structures generally support more efficient and cooperative team dynamics. These findings are particularly relevant in the NHL, where total spending is capped,

but salary distribution within teams is flexible. As general managers construct rosters, they must consider not only marginal productivity but also the interpersonal and strategic costs of wage imbalances. Fairness, in this context, becomes a driver of on-ice performance.

2.3.4 Contract Structure and Incentive Design

The structure of player contracts, shaped by collective bargaining agreements, salary caps, and free agency rules, strongly influences incentive effects. Frick (2011) shows that even guaranteed contracts can produce strong performance incentives, particularly in contract years, due to complementary institutional mechanisms such as arbitration and transfer markets. His analysis contrasts European soccer, which lacks a hard salary cap, with North American leagues like the NHL and NFL. In uncapped leagues, there is often tighter alignment between a player's marginal revenue product and compensation, allowing standout performance to be directly rewarded. However, this also increases the risk of overpayment and can create performance pressures at critical points in a player's contract cycle.

Contract duration plays a dual role in shaping incentives. While long-term deals offer financial security, they also increase the risk of shirking and performance volatility. To mitigate these risks, teams may include performance triggers, front-loaded salary structures, or opt for shorter contracts. In the NHL, where salaries are guaranteed but heavily scrutinized under a hard cap, teams often weigh these trade-offs carefully when managing risk and reward.

2.3.5 Application to the NHL Context

The existing literature suggests that contract year effects are measurable and meaningful, though the strength and nature of the effect vary by league and institutional design. The NHL's distinctive features, a hard salary cap, fully guaranteed contracts, and relatively limited use of performance-based incentives, create a unique environment for testing how short-term financial incentives shape behavior.

From a methodological standpoint, the literature supports our empirical strategy. Specifically,

testing contract year effects using panel regression with player-fixed effects helps control for unobserved, time-invariant differences in talent, role, or playing style. Additionally, incorporating lagged performance variables helps reduce simultaneity bias and better isolates the effect of contract timing from broader performance trends.

Ultimately, the contract year effect is not just a narrative pushed by media or fans, it reflects a real economic question about how players respond to incentives under constrained conditions. Our analysis contributes to this broader conversation by focusing on the NHL, a setting where rigid contract structures and team-level constraints may dampen or alter the typical patterns found in other leagues.

3 Methodology

This section outlines the empirical strategy used to examine the effect of contract year status on NHL player performance. Specifically, we estimate whether players score more or fewer points in seasons when they are in the final year of their contract. We use a fixed effects panel regression model to account for both player-specific attributes and league-wide seasonal effects.

3.1 Explanation of Model: Skaters

Our goal is to test if NHL skaters perform better in the final year of their contract. To do this, we run a fixed effects panel regression with points per game (PPG) as the dependent variable. PPG is calculated by dividing a player's total points by the number of games they played that season. This helps us control for differences in ice time, injuries, or short seasons.

The main variable we're focused on is a contract year dummy. It equals 1 if the player is in the last year of their current deal, and 0 otherwise. We want to see if players boost their performance when they're playing for their next contract. A common though among professional athletes.

We also include age and age squared to account for the fact that performance usually follows a curve—most players improve early in their careers, peak in their mid-to-late 20s, and then decline.

Including both terms helps us capture that full arc.

To control for a player's offensive role and ability, we include lagged stats from the previous season: lagged points per game, lagged expected goals per game, and lagged primary assists per game. These help us adjust for how good a player is going into the season, without using stats from the current year that could be affected by the contract year itself.

We're using player fixed effects to compare each player only to themselves over time. This controls for anything about the player that doesn't change from year to year—like position, skill level, or style of play. We also include year fixed effects to control for stuff that affects the whole league, like rule changes, lockouts, or shifts in scoring trends. Standard errors are clustered at the player level since seasons for the same player aren't independent.

This setup follows what other papers in the contract year literature have done. For example, Stiroh (2007) and Maxcy et al. (2002) both use lagged stats to account for past performance and avoid reverse causality, which is what we're doing here too.

Here's what the model looks like:

$$\text{PPG}it = \beta_0 + \beta_1 \cdot \text{ContractYear}it + \rho$$

$$+\beta_4 \cdot \text{LaggedPPG}i,t-1 + \beta_5 \cdot \text{LaggedxGPG}i,t-1 + \beta_6 \cdot \text{LaggedPrimaryAstPG}i,t-1 \; + \delta_t + \alpha_i + \varepsilon it \tag{1}$$

We also split the data to run the same model separately for unrestricted free agents (UFAs) and restricted free agents (RFAs). UFAs are usually older players who can sign with any team, while RFAs are still under team control. This lets us test if players respond differently depending on how much leverage they have.

Those two models are the same as above, just limited to UFA and RFA groups:

$$\text{PPG}_{it}^{\text{UFA}} = \text{Same model, but only includes players where is}_u fa \text{is}_u fa \text{is}_{u fa \text{is}_u fa = 1(2)}$$

$$\text{PPG}_{it}^{\text{RFA}} = \text{Same model, but only includes players where is}_{r}fa\text{is}_{r}fa\text{is}_{r}fa_{\text{is}_{r}fa=1(3)}$$

All three versions use the same controls and setup, so we can compare the results side by side and see if contract year effects look different depending on a player's free agency status

3.2 Explanation of Model Goalies

To estimate the effect of contract years on goalie performance, we use a fixed effects panel regression model. This method helps us focus on changes in performance within each goalie over time. It controls for all of the personal traits that do not change from year to year, such as natural talent, playing style, or the role the goalie usually plays on their team. By looking at how the same goalie performs in different seasons, we can better understand whether they play better or worse during a contract year.

We also include season fixed effects to control for things that might affect all goalies in a given year. This includes changes in league scoring patterns, rule updates, schedule changes, or other shifts that could make the league easier or harder for goalies. These year effects help make sure our results are not affected by changes that have nothing to do with individual performance.

To make the model even stronger, we add the goalie's performance from the previous season as a control. This helps account for performance trends and reduces the risk that we are picking up effects that go the other way, like goalies getting contract years because they already played well. We also control for the number of games each goalie played and how they performed on high danger shots. This helps us adjust for differences in playing time and shot quality. With all of these factors included, we can better measure how much contract years impact how goalies play.

$$\begin{aligned} \text{GSAX}_{it} &= \beta_0 + \beta_1 \cdot \text{ContractYear}_{it} + \beta_2 \cdot \text{GSAX}_{i,t-1} \\ &+ \beta_3 \cdot \text{GamesPlayed}_{it} + \beta_4 \cdot \text{HighDangerGSAX}_{it} + \delta_t + \alpha_i + \varepsilon_{it} \end{aligned} \tag{4}$$

3.3 Explanation of Model: Goalies

To estimate the effect of contract year status on goalie performance, we use a panel regression model with season fixed effects and lagged performance controls, but without individual fixed effects. This specification allows us to assess both within-player changes and differences across players, enabling a broader evaluation of whether contract status systematically affects performance. We include season fixed effects to account for league-wide trends — such as rule changes, shifts in scoring environments, or scheduling differences — that could affect all goalies in a given season. This helps ensure that our estimated effects are not confounded by structural changes in the league context. To mitigate potential issues of reverse causality, where prior performance might influence contract timing, we use lagged values of key performance indicators. This ensures that all explanatory variables reflect performance from the previous season, isolating the impact of entering a contract year on current-season outcomes. The dependent variable in our regression is Goals Saved Above Expected (GSAx), a modern, shot-quality adjusted performance metric that evaluates a goaltender's effectiveness more comprehensively than traditional statistics like save percentage or goals-against average. GSAx compares the number of goals a goalie actually allowed to the number of goals they were expected to allow based on detailed shot characteristics—such as shot location, shot type (wrist, slap, deflection, rebound, etc.), pre-shot movement, distance from the net, and whether the shot came off a rush or power play. This metric is designed to isolate goalie performance from team defensive quality. A positive GSAx value indicates that the goalie prevented more goals than an average NHL goalie would have, given the same shot profile, while a negative value suggests underperformance relative to expectations. We chose GSAx over traditional metrics like save percentage because save percentage treats all shots equally, regardless of quality. A save on a routine shot from the blue line is weighted the same as a breakaway or one-timer from the slot. This makes save percentage highly sensitive to team context: a goalie behind a strong defensive team might face easier shots and therefore post a deceptively high save percentage, while a goalie behind a weaker defense may appear worse despite facing more difficult chances. GSAx corrects for this by accounting for shot quality, providing a more nuanced

and fair evaluation of individual goalie performance. In addition to overall GSAx, we also include lagged high-danger GSAx, a more focused metric that captures how goalies perform specifically against high-danger scoring chances. These are shots taken from areas with a significantly higher likelihood of resulting in a goal—usually defined as shots from the slot or crease area, within close proximity to the net, and often involving lateral puck movement. High-danger GSAx isolates a goalie's ability to handle the most threatening opportunities, which are often decisive in close games and are more reflective of a goalie's skill and reflexes under pressure. This variable allows us to evaluate not just general shot-stopping ability, but how well goalies perform in the most critical moments. Finally, we include lagged games played as a control for prior-season workload. This variable serves as a proxy for both team confidence in the goalie and physical fatigue, while also helping to differentiate starters from backups. By using these lagged and shot-quality adjusted variables, our model aims to offer a more accurate and robust understanding of how goalie performance responds to contract incentives. The regression equation is specified as follows:

$$\begin{aligned} \text{GSAX}_{it} &= \beta_0 + \beta_1 \cdot \text{ContractYear}_{it} + \beta_2 \cdot \text{GSAX}_{i,t-1} \\ &+ \beta_3 \cdot \text{GamesPlayed}_{it} + \beta_4 \cdot \text{HighDangerGSAX}_{it} + \delta_t + \alpha_i + \varepsilon_{it} \end{aligned} \tag{5}$$

Where:

- $GSAx_{it}$: Goalie i's goals saved above expected in season t
- $ContractYear_{it}$: Indicator for the final year of goalie i's contract
- $GSAx_{i,t-1}$: Prior-season performance, adjusted for shot quality
- $GamesPlayed_{i,t-1}$: Prior-season workload
- $HighDangerGSAx_{i,t-1}$: Prior-season performance on high-danger shots
- δ_t: Season fixed effects
- ε_{it}: Error term

4 Description

4.1 Data Collection

For this paper, we collected player statistics and contract data from three primary sources, covering the 2004–05 to 2024–25 NHL seasons. This time frame was selected because it corresponds to the Salary Cap Era, which fundamentally altered the league's salary structure. First, we obtained basic player and team statistics using the NHL's open-source API. To access this data, we utilized an unofficial Python wrapper developed by Zmalski, available in the NHL-API-Reference repository on GitHub, which facilitates the retrieval of season-by-season team and player data. Second, we used MoneyPuck, a public analytics resource that provides advanced player metrics from the 2008–09 season to the present. The data is organized in individual CSV files by season, with separate files for skaters and goaltenders. Lastly, contract information was sourced from Spotrac. This dataset was purchased for a fee of $30 and provided in team-by-team CSV files.

To prepare our dataset for analysis, we wrote a series of Python scripts to clean, expand, and merge salary and performance data across multiple sources. First, we used expanded multi-year contracts from Spotrac so that each season of a player's deal became its own observation. For example, a five year contract starting in 2023 was split into five rows, one for each year of the contract, with the correct season assigned to each row. We then created a player level panel that included a breakdown of contract year, player age, and an indicator variable identifying whether the observation represented the final year of a contract. This variable was of a boolean type.

After this, we merged the expanded contract data with a base performance dataset containing yearly statistics such as goals, assists, and games played. This required cleaning and aligning player names and transforming season strings into usable year formats. To build the dataset further, we merged in advanced skater statistics from MoneyPuck, again using player name and year as join keys with proper string cleaning. This set our data to be loaded into state where we could do further data cleaning.

4.2 Data Description: Skaters

Some of the stats we used in our dataset are pretty basic things like goals, assists, and games played don't really need explaining. But a few of the others are a bit more advanced, and since they're a big part of our model, we figured it was worth breaking them down.

One of the main ones is expected goals. This tells us how many goals a player should've scored based on shot quality. It looks at things like where the shot was taken from, the angle, the type of shot, and a few other things. Not every shot is created equal a quick rebound in tight is much more dangerous than a wrister from the blue line and expected goals helps capture that. It gives a better idea of how dangerous a player actually was, rather than just looking at how many goals they happened to score.

We also used primary assists, which only count the last pass before a goal. This is important because it tells us who actually made the key play that directly led to the goal. In the NHL, up to two assists can be awarded on a single goal—the primary and the secondary. But the secondary assist is just the second-to-last player to touch the puck, and sometimes it's not really related to the actual play that created the goal. It could be a simple breakout pass or a lucky bounce. So while it still counts on the scoresheet, it doesn't always reflect real playmaking. That's why we focus on primary assists. They give us a better sense of who's setting up goals and making smart, creative plays in the offensive zone, not just racking up points from being in the right place at the right time.

Giveaways are another stat we included. These are just how often a player gives the puck away. A high number of giveaways might mean a guy is careless, or maybe just trying to force plays that aren't there. Either way, it's not great, and it's something teams look at when figuring out value.

To better track shot quality, we also added in some high danger stats: high-danger shots, high-danger expected goals, and high danger goals. These are the best chances in the game the ones where the goalie barely has time to react. When guys are generating these chances consistently, it shows they're actually creating real offense, not just throwing pucks on net from bad angles.

We looked at even strength points too, which are points scored during 5-on-5 play. This helps

level things out a bit, since some guys rack up power play points and others don't get much special teams time. Even-strength production is a good way to compare guys more fairly.

One of the key variables in our model is the contract year dummy. It's a 1 if the player is in the final year of their current deal, 0 otherwise. That's what we use to test if guys step it up in contract years. About 41.1% of player-seasons in our data were contract years. We also added dummy variables for RFA and UFA status, using the same idea.

On top of that, we tossed in a few more stats for context. Game Score is a single number that sums up a player's impact in a game it looks at things like goals, assists, shots, and a bunch of other stuff. Plus-minus shows how many goals were scored for and against while a player was on the ice. It's kind of flawed, but it still gets used a lot. Points per game is helpful when guys miss time it gives you a rate stat instead of a raw total.

Finally, we used per game versions of a few key stats points per game, xG per game, and primary assists per game. This helps us make fair comparisons when a player misses time because of injuries, scratches, or shortened seasons like during COVID. Per-game stats keep everyone on the same playing field.

4.3 Data Cleaning Skaters

We used a combination of Python and Stata to clean and merge our datasets. Observations from seasons before 2008–09 were excluded due to limitations in the coverage of the MoneyPuck dataset. For the contract data, we expanded each player's contract into individual yearly observations for every season the contract was active. We then merged the datasets using Python's pandas library, joining on a composite key consisting of the player's full name and the season. The final dataset for skaters contains 10,347 observations of data. After reviewing the summary statistics of the skaters' dataset, including the mean, standard deviation, min, and max, we determined that no significant outliers required removal. We did run into concerning Max stat. This was in games played where the max is 108. Given that NHL only plays 82 games this was a problem we thought. After further investigation we found that the games played variable accounts for both

regular season and playoff games that played. A summary of the dataset is presented in Table. 1 below.

Table 1: Summary Statistics for NHL Skaters (2008–2024)

Variable	**Obs.**	**Mean**	**Std. Dev.**	**Min**	**Max**
Year	10,347	2016.49	4.71	2008	2024
Age	10,347	25.73	4.35	17	44
Age^2	10,347	680.83	234.65	289	1936
AAV	10,347	3,158,041	2,618,217	92,500	14,000,000
Contract Year Dummy	10,347	0.411	0.492	0	1
UFA Dummy	10,347	0.397	0.489	0	1
RFA Dummy	10,347	0.603	0.489	0	1
Goals	10,347	10.10	10.10	0	72
Assists	10,347	16.99	14.69	0	134
Points	10,347	27.09	23.19	0	174
Points per Game	10,347	0.418	0.261	0	3
Expected Goals	10,347	9.63	8.20	0	54.65
Primary Assists	10,347	9.10	8.10	0	62
Individual Points	10,347	26.20	21.19	0	153
Individual Goals	10,347	9.78	9.36	0	69
Giveaways	10,347	28.47	21.18	0	159
High Danger Shots	10,347	9.37	9.68	0	81
High Danger xGoals	10,347	3.30	3.58	0	28.08
Lagged Points	8,078	30.70	23.57	0	174
Lagged xGoals	8,078	10.63	8.26	0	54.65
Lagged Primary Assists	8,078	10.14	8.21	0	62
Lagged PPG	8,078	0.449	0.263	0	3
Lagged xGoals per Game	8,078	0.165	0.139	0	5.46
Lagged Primary Assists per Game	8,078	0.156	0.144	0	7

We should note that a significant number of variables from the original dataset, particularly those sourced from the MoneyPuck database, were excluded during the data cleaning process. This decision was driven by both methodological and conceptual considerations related to the goals of our study.

First, we removed statistics related to low-danger scoring opportunities. These variables capture events such as long-range, low-probability shots that rarely lead to goals. While useful in some contexts, we found that these metrics largely reflect poor shot quality or inefficiency, which contrasts with our focus on identifying player overperformance. Since our study aims to assess

how players generate high-quality outcomes, particularly with contract year incentives, low-danger statistics were not aligned with our objectives.

Next, we dropped several defensive-related metrics, including blocked shots, hits, and faceoff statistics. Although these are undeniably important aspects of hockey performance, they are typically associated with specific player roles, such as defensive specialists or fourth-line grinders, who tend to be on smaller contracts and contribute in ways not directly tied to scoring output. Because our primary interest lies in the offensive production and contract-related behavior of players likely to influence salary negotiations, we determined that such role-specific stats were not central to our research question.

We also removed a category of variables that track on-ice team performance while the player is on the ice, such as expected goals for or against, Corsi and Fenwick percentages, and shot attempt differentials. These are commonly used in advanced analytics to assess a player's two-way impact within a team structure. However, they are heavily influenced by linemates, team systems, and coaching strategies. While we acknowledge their value in broader hockey analysis, they were not directly relevant to our question of how individual performance, particularly scoring efficiency and offensive contribution, varies in a contract year. Overall, our feature reduction was guided by a desire to focus the model on individual, offense-driven performance metrics that are both interpretable and central to player valuation, especially in the context of contract negotiations and free agency behavior.

Finally, while preparing the panel dataset for regression, we encountered an error indicating that some time periods contained multiple observations for the same player ID. Upon investigation, we discovered that this issue arose from players who were traded mid-season, leading to multiple entries for the same individual within a single season. To resolve this, we removed the player's second appearance during that season, thereby ensuring a unique ID player season structure in the panel. We do not believe this adjustment significantly biases our results. Players who are traded mid-season typically do not command high-value contracts, and our analysis is primarily focused on more prominent skaters who are eligible for high-salary contracts. As well, by removing the

second appearance, we are removing their smaller sample as the trade deadline is later in the season.

4.4 Data Description Goalie

The final dataset includes 635 NHL goalie-seasons from 2009 to 2024, with one row per player-season. This structure allows us to analyze how prior-season performance metrics relate to current-season outcomes in the context of contract status. The outcome variable is Goals Saved Above Expected (GSAx), a shot-quality adjusted metric that captures goaltender performance independent of defensive team effects. The explanatory variables include a binary indicator for ContractYear, along with lagged GSAx, lagged high-danger GSAx, and lagged games played. These controls are chosen to account for previous skill level, shot difficulty, and workload. All variables are measured consistently across seasons and formatted for use in panel regression with season fixed effects. This dataset enables us to examine whether goaltenders respond strategically to contract incentives, while accounting for baseline ability and broader league trends. Descriptive statistics summarizing the variation in performance and playing time are presented in Table 2.

4.5 Data Cleaning Goalie

We followed a similar data collection process as outlined in the previous section, with one notable exception: the goalie dataset does not include data from the NHL API. Instead, we relied exclusively on statistics from MoneyPuck. These data were merged using each goaltender's name and season to create a panel of season level observations. After merging, we removed situation specific breakdowns and retained only the rows labeled "all," which summarize total performance across all game states. This produced an initial dataset of 1,408 goalie-season observations, averaging approximately 70 per season. This is consistent with each NHL team typically employing two primary goaltenders per season—one starter and one backup—with occasional turnover due to trades or injuries. To prepare the data for regression analysis, we filtered out all goalie seasons with 24 or fewer games played. This threshold was chosen to exclude backups and emergency call ups whose

limited ice time could bias performance metrics. We further ensured that each goalie season was represented by a single observation by eliminating residual duplicates that arose from situation specific entries. To construct a lagged measure of performance, we defined the panel structure using xtset with playerid and season as the identifiers. We then generated a one-year lag of Goals Saved Above Expected (GSAx). Observations without a valid lagged value, typically corresponding to a goalie's first season, were removed. The resulting cleaned dataset includes 635 goalie-seasons spanning from 2009 to 2024. Goalies in the sample played an average of 45.4 games per season (minimum: 25, maximum: 77), with a mean GSAx of 1.82 and a standard deviation of 12.86, suggesting considerable variation in performance. On average, goalies faced 1,296 shots, with 116 expected goals and 114 actual goals allowed per season. Notably, average high danger expected goals allowed was 35.84, while actual high-danger goals allowed was slightly lower at 34.06 indicating that goaltenders, on average, slightly exceeded expectations in these critical situations. The binary contractyear variable has a mean of 0.40, indicating that approximately 40

Table 2: Summary Statistics for Goalie Season-Level Data (n = 635)

Variable	Obs	Mean	Std. Dev.	Min	Max
Player ID	635	8,473,034	4,730.64	8,455,710	8,481,611
Season Year	635	2016.58	4.66	2009	2024
Games Played	635	45.39	13.30	25	77
Time on Ice (Seconds)	635	154,874.5	48,607.05	69,021	267,541
Expected Goals Allowed	635	116.07	36.84	47.14	217.67
Goals Allowed	635	114.25	34.24	43	200
Unblocked Shots Faced	635	2,427.62	770.23	1,032	4,450
Expected Rebounds	635	89.16	28.27	36.67	168.82
Rebounds Allowed	635	100.00	37.54	24	222
Expected Freezes	635	299.32	94.83	121.54	555.09
Freezes	635	290.38	100.64	106	575
Expected Shots on Goal	635	1,299.10	411.00	538.09	2,458.88
Shots on Goal	635	1,296.34	412.69	530	2,375
Expected Stoppages of Play	635	41.83	13.75	16.36	73.21
Stoppages of Play	635	38.29	21.85	0	146
Expected Continued Play (Neutral Zone)	635	722.31	228.99	296.98	1,344.56
Continued Play (Neutral Zone)	635	581.61	192.12	234	1,104
Expected Continued Play (Defensive Zone)	635	536.44	171.95	222.85	1,030.48
Continued Play (Defensive Zone)	635	682.65	220.66	270	1,324
Flurry-Adjusted Shots	635	111.33	35.19	45.69	207.07
Low-Danger Shots Faced	635	1,359.38	439.68	546	2,581
Medium-Danger Shots Faced	635	339.71	107.22	153	683
High-Danger Shots Faced	635	108.83	38.65	28	228
Low-Danger xGoals	635	38.94	12.37	15.96	75.81
Medium-Danger xGoals	635	41.29	13.09	18.53	83.57
High-Danger xGoals	635	35.84	12.91	9.06	71.46
Low-Danger Goals Allowed	635	39.38	13.16	8	77
Medium-Danger Goals Allowed	635	40.81	12.88	12	91
High-Danger Goals Allowed	635	34.06	12.37	8	75
Blocked Shots	635	619.69	208.16	232	1,163
Penalty Minutes	635	2.43	3.12	0	25
Penalties Taken	635	1.20	1.48	0	8
Years Remaining on Contract	635	2.41	1.79	1	12
Contract Year Indicator	635	0.40	0.49	0	1
Goals Saved Above Expected (GSAx)	635	1.82	12.86	-40.89	46.67
Save Percentage	635	0.911	0.011	0.871	0.940
Save Percentage (Percent Format)	635	91.07	1.13	87.11	94.03
High-Danger GSAx	635	1.78	7.14	-25.26	31.88
Lagged GSAx	635	2.09	11.47	-35.18	46.67

5 Results

5.1 Skaters Results

Table 3: Fixed Effects Regression — Contract Year and Lagged Performance (All, UFA, RFA)

Variable	All Skaters	UFA Skaters	RFA Skaters
Contract Year	-0.001	0.005	-0.000
	(0.003)	(0.005)	(0.005)
Age	0.089***	0.098***	0.183***
	(0.011)	(0.028)	(0.031)
Age^2	-0.002***	-0.002***	-0.004***
	(0.000)	(0.000)	(0.001)
Games Played	0.001***	0.001***	0.001***
	(0.000)	(0.000)	(0.000)
Lagged PPG	0.209***	0.082***	0.106***
	(0.020)	(0.026)	(0.023)
Lagged xGoals per Game	0.093***	0.077*	0.086**
	(0.030)	(0.043)	(0.041)
Lagged Primary Assists per Game	-0.117***	-0.087**	-0.100**
	(0.032)	(0.042)	(0.039)
Year Fixed Effects	Yes	Yes	Yes
Player Fixed Effects	Yes	Yes	Yes
Observations	8,078	3,718	4,360
Players (Groups)	1,432	881	1,204
R^2 (Within)	0.205	0.268	0.118

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors clustered at the player level.

We ran our regression using Equation 1 from Section III to estimate whether NHL skaters increase their performance during the final year of their contracts. Table 3 shows the results for all players, as well as separately for UFAs and RFAs. Our main takeaway is the following, across all groups, there is no evidence of a contract year effect. The coefficients on the contract year variable are near zero and statistically insignificant in every case measured, this suggests that players do not change their behavior in contract years at least in terms of offensive output it does not change.

This finding contradicts a popular belief in sports that players elevate their game when it's time to earn a new deal. But when viewed through the structure of the NHL labor market, the result makes a lot of sense. Unlike in the NBA or MLB, where players can see massive salary jumps

from free agency and performance bonuses, NHL contracts are fully guaranteed and often smaller in size when compared to other major sports leagues. Players are paid whether they meet certain performance milestones or not, and large incentive based bonuses are rare outside of entry-level contracts or special one year deals for veterans. That limits the upside of a big season.

The league's hard salary cap plays a major role here, too. Even if a player performs well in their contract year, their earning potential is capped by team budgets. Many teams are operating right up against the ceiling, which reduces bidding wars and the reward for marginal gains in performance. That also means teams are more cautious about handing out large contracts based on one good season, instead preferring consistent, multiv year performance.

Another key contribution of our model is the inclusion of lagged performance controls, such as prior season points per game, expected goals, and primary assists. These variables help isolate the effect of being in a contract year by accounting for a player's recent performance. In our results, these lagged stats are consistently significant, reinforcing the idea supported by studies like Stiroh (2007) and Maxcy et al. (2002) that past performance is a strong predictor of both current on-ice/field performance. However, while those papers also find evidence of a contract year boost, our findings differ; once we control for past performance, the contract year effect is not longer statistically significant. This suggests that what might look like a performance bump in some cases could just be a continuation of recent trends, rather than a true response to the prospects of earning more money in a new contract

We also control for player age and age squared, which capture career trajectory. We find that age has a nonlinear relationship with performance players improve in their early to mid-20s before peaking and gradually declining. Interestingly, when we split the sample into UFAs and RFAs, we see that RFAs generally younger players have a steeper age-performance curve, as expected. But again, contract year status doesn't significantly impact performance in either group.

The use of fixed effects is another strength of our model. By comparing each player to themselves over time, we control for individual characteristics that don't change like position, play style, and baseline talent. This rules out the idea that only certain types of players behave differently in

contract years. Even within the same player, across different seasons, we still don't see a bump in performance tied to contract timing. Year fixed effects also ensure that league wide changes like COVID seasons or rule changes don't bias the results.

Some contract-year players may be used less or given fewer opportunities maybe their coaches know they won't be re-signed. But if that were common and meaningful, we would expect to see lower performance in those years. We don't. The coefficient on contract year is basically zero. So whatever roster decisions are being made behind the scenes, they don't appear to affect the player's offensive output.

To assess the importance of controlling for unobserved player heterogeneity, we estimate random effects models using the same set of lagged performance controls and year fixed effects Table 4. In the full sample, the contract year coefficient is negative (0.007) and marginally significant at the 10% level, suggesting a small decline in points per game during contract years when player fixed effects are omitted. This effect, however, disappears in the fixed effects model Table 3, indicating that unobserved player traits confound the simple relationship between contract year status and performance. The UFA and RFA subsample results show similar patterns: the contract year coefficients are negative but statistically insignificant in both cases. These findings support our preferred fixed effects specification and emphasize the need to control for player-specific factors when estimating contract year effects

Table 4: Random Effects GLS Regression — Contract Year and Lagged Performance (All, UFA, RFA)

Variable	**All Skaters**	**UFA Skaters**	**RFA Skaters**
Contract Year	-0.007*	-0.006	-0.008
	(0.004)	(0.005)	(0.005)
Age	-0.006	0.003	-0.037
	(0.004)	(0.015)	(0.026)
Age^2	0.000	-0.000	0.001
	(0.000)	(0.000)	(0.001)
Games Played	0.001***	0.001***	0.002***
	(0.000)	(0.000)	(0.000)
Lagged PPG	0.725***	0.716***	0.779***
	(0.010)	(0.014)	(0.013)
Lagged xGoals per Game	0.211***	0.183***	0.228***
	(0.021)	(0.028)	(0.029)
Lagged Primary Assists per Game	-0.207***	-0.115***	-0.246***
	(0.021)	(0.034)	(0.027)
Year Fixed Effects	Yes	Yes	Yes
Player Effects	Random	Random	Random
Observations	8,078	3,718	4,360
Players (Groups)	1,432	881	1,204
R^2 (Within)	0.129	0.121	0.044

Notes: *** p<0.01, ** p<0.05, * p<0.1.

In summary, our results suggest that NHL skaters do not perform differently in contract years, regardless of whether they are UFAs or RFAs. This challenges the widespread assumption that athletes elevate their play when money is on the line. In the NHL's system of guaranteed contracts, hard salary caps, and relatively flat market incentives, the data tells a story that reads as follows, most players just keep doing what they always do. That might not be the sports narrative we're used to seeing in but it's what the numbers show.

5.2 Goalie Results

Our updated model examines the relationship between contract year status and goalie performance using a panel regression with lagged controls and season fixed effects. By using lagged perfor-

mance metrics, we aim to reduce concerns about reverse causality—specifically, that a goalie might enter a contract year following strong prior-season performance. The regression results show that the coefficient on contract year is negative but statistically insignificant, suggesting that goalies do not experience a measurable change in performance during their contract year. This finding stands in contrast to the commonly held belief that players elevate their performance in contract seasons to secure more favorable deals. Instead, our results imply that goalie performance may not be strongly influenced by contract incentives, or that teams may already factor in multi-year trends and situational context when evaluating goalies for new contracts. Among the control variables, lagged GSAx is negatively and insignificantly associated with current-season GSAx, suggesting that past performance does not consistently predict future outcomes for goaltenders. Notably, lagged games played is also negatively related to current GSAx, indicating that goalies who logged more playing time in the prior season might experience diminished performance—perhaps due to fatigue or other adverse factors—although this relationship remains only marginally insignificant. Additionally, lagged high-danger GSAx continues to show a weak and statistically insignificant relationship with current GSAx outcomes, implying that performance against high-danger shots in one season does not reliably translate into overall performance the following year. Overall, these results suggest that contract year status does not significantly affect goalie performance when controlling for past performance, workload, and shot difficulty. This may be due to the unique nature of the goalie position, where consistency, coaching systems, and team defensive structure play larger roles, or because teams place less weight on contract-year performance when making goaltending decisions. Additionally, goalies are very unpredictable and often face circumstances beyond their control—ranging from the variability in defensive coverage to unpredictable game situations—which means they may not always be fully in command of their performance. This inherent unpredictability further obscures any systematic effects of contract year dynamics on performance.

Table 5: Fixed Effects Regression Results: Goalie GSAx

Variable	Coefficient	Std. Err.	t	p-value
Contract Year	-0.2033	1.2668	-0.16	0.873
Lagged GSAx	-0.0315	0.0831	-0.38	0.705
Lagged Games Played	-0.1181	0.0643	-1.84	0.069
Lagged High Danger GSAx	0.0031	0.1561	0.02	0.984
Constant	12.7163	4.5448	2.80	0.006

6 Closing Thoughts

This paper set out to test the widely held belief that NHL players improve their performance during contract years. Drawing on a rich panel dataset from 2008 to 2024, we used a fixed effects regression model to isolate player changes in performance, controlling for age, experience, lagged productivity, and league-wide trends. Contrary to what is often found in other major sports leagues, such as the NBA or MLB, our results show that NHL skaters tend to perform worse, not better, during contract years, which was unexpected. The coefficient on the contract year variable is statistically significant and negative, even after controlling for known drivers of offensive output like expected goals, primary assists, and total games played.

These findings are both surprising and telling. They challenge the conventional narrative that players elevate their performance when new contracts are at stake. Instead, the NHL's structural conditions, particularly its hard salary cap, the prevalence of long-term guaranteed deals, and cautious front-office behavior, may foster an environment in which contract years do not act as strong individual motivators. Players may have reasons to play more casually to avoid injury and protect future earnings potential.

Our findings also speak to the unique lifecycle dynamics of NHL players. Contract years may occur at developmentally awkward moments, either at the tail end of rookie contracts when players are still maturing, or at the end of careers, when performance is naturally declining. Despite including nonlinear controls for age and player fixed effects, contract year status may still capture some of this variation, particularly if it affects deployment decisions like ice time or special team opportunities.

More broadly, this research contributes to a growing literature in sports economics that explores how labor market structures influence individual incentives. It underscores that the contract year effect is not universal, but rather dependent on institutional context. In a league like the NHL where salaries are capped, most contracts are guaranteed with minimal player performance incentives built into the contract, and team systems often dictate on-ice roles, player behavior appears less sensitive to short-term contractual incentives.

Looking ahead, future research could extend this work in several directions. One promising path would be to explore post-contract performance to test for shirking effects and long-run value delivery. Another could focus on subgroups, such as high-salary players, entry-level contract holders, or those negotiating through arbitration, to see whether the contract year effect varies by role or bargaining position.

Ultimately, while our results complicate the standard contract year narrative, they provide a valuable reminder that incentives do matter, but how they matter depends on the structure of the system they operate. In the case of the NHL, contract years may not inspire heightened performance, but rather more cautious and strategic play.

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